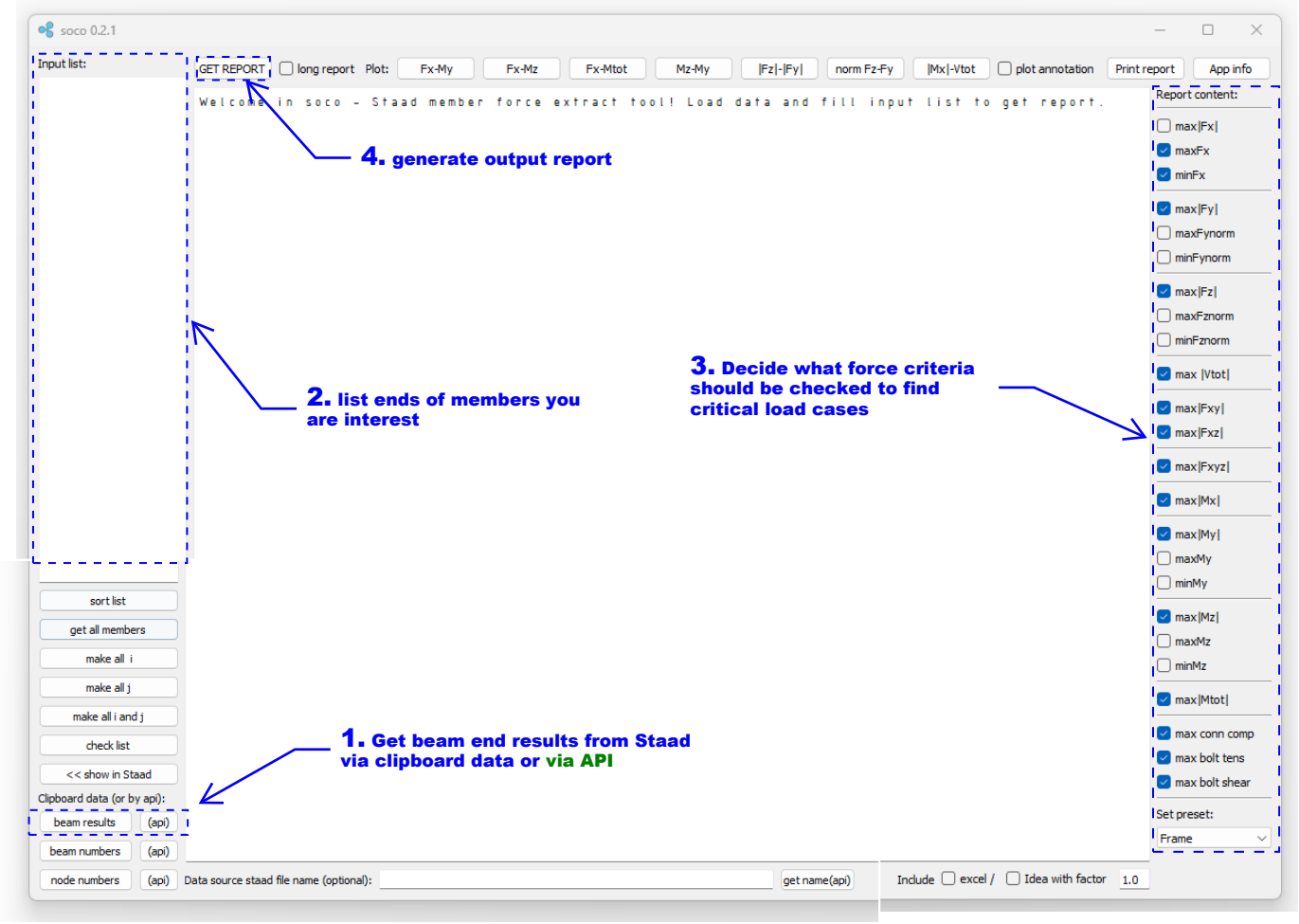
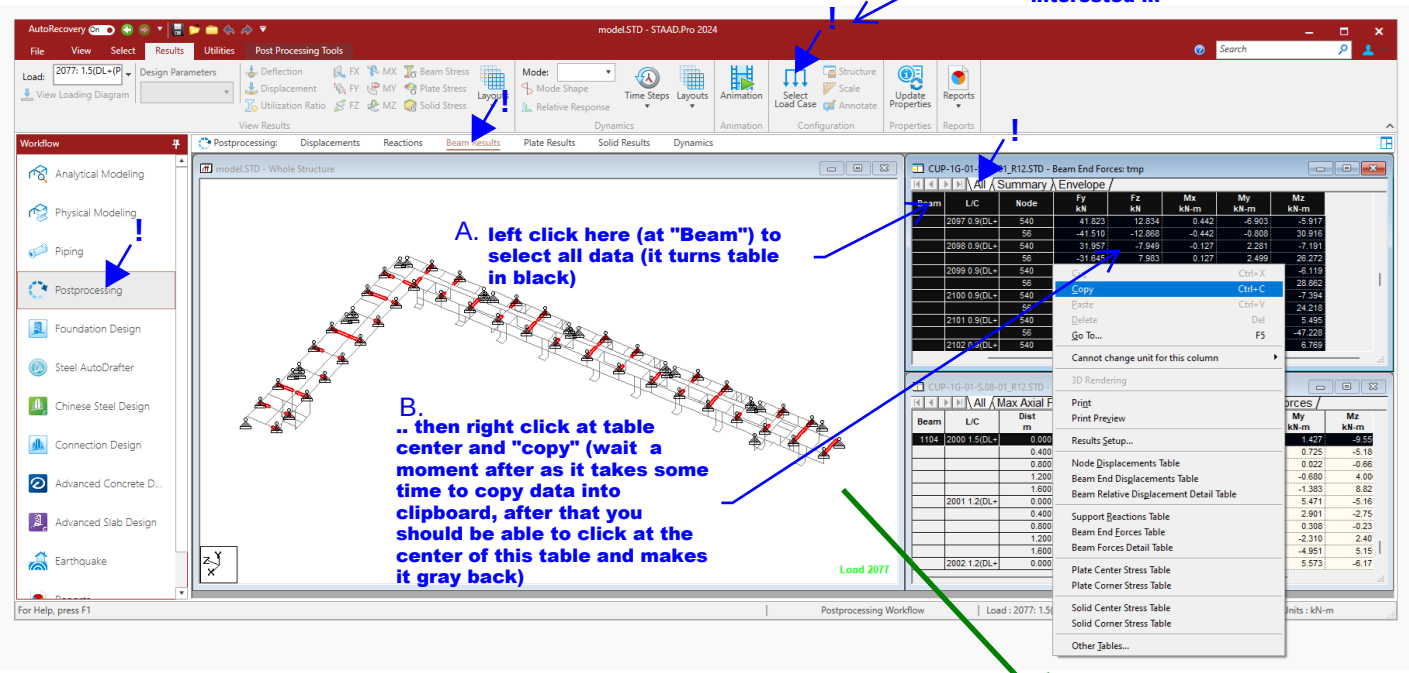
Basic workflow steps (each step explained in details on next pages)



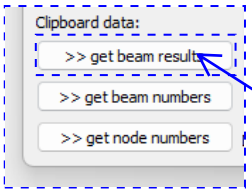
1. Geting beam end results from Staad(*)

Note - here decide what load combinations you are interested in



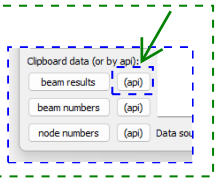
(*) Note. Currently data transfer from Staad is possible via clipboard by coping Staad tables (the old method), **or by using Staad API** (new method in version 0.3.1). The old method is a bit not handy and because of "Range too big" issue working with big Staad models is more difficult as you can not transfer entire model results at once. New API method solve this issue.

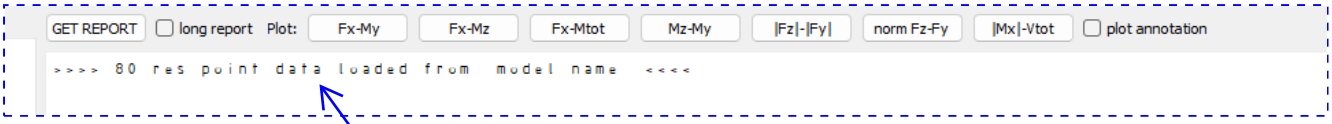
C. go to SOCO



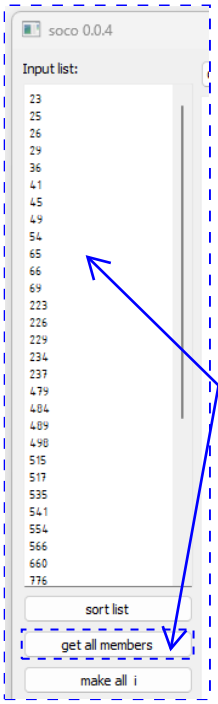
D. **click on "get beam results" button**

New API method - just select members and click (if no member selected results for all member will be taken)



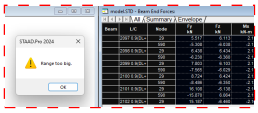


E. **you should see confirmation that member results has been received by SOCO**



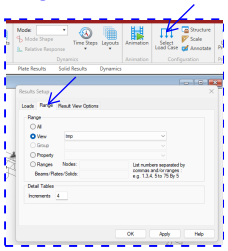
F. **you can use "get all members" button to see what member results are available**

"Range too big." issue (when using data transfer via clipboard).

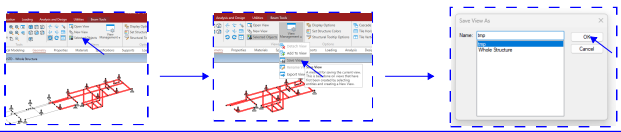


(Note - Use API data transfer instead to solve this problem.)

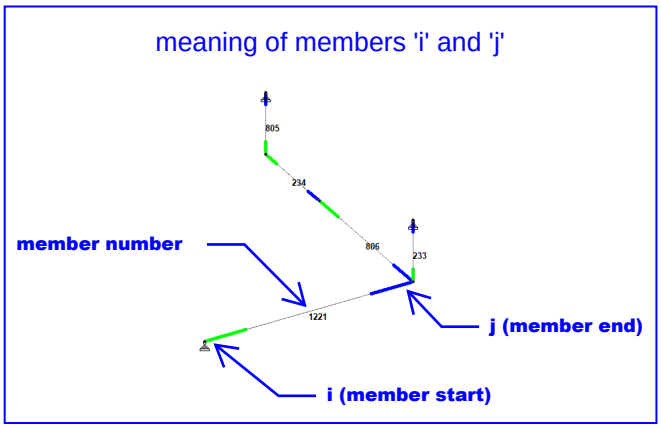
For big models or/and many load combinations results table can be to big to copy all members results for entire. For that case you need to reduce the table size by using some criteria as shown below.



Using 'View' option is most flexible as you can define views in fly with no impact on std file and no need recalculate model when create new view. See below how to crate new named view in Staad.

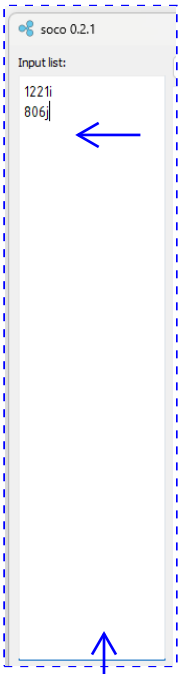


2. Listing ends of members you are interested in



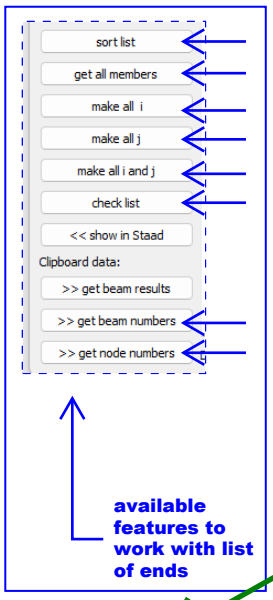
There are two options how to specify list of ends

Option 1. Just write it manually



List of ends need to be specified here

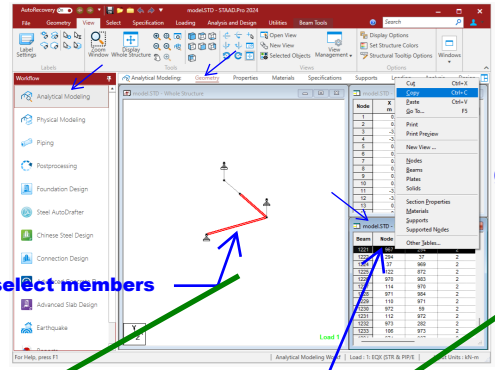
Option 2. Use Soco build in features that make it easer to create and menage the list



available features to work with list of ends

Specifying list of ends based on Staad member and point selection is shown below.

Fist get beam numbers based on Staad selection:



1. select members

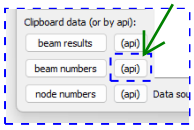
2. right click here and copy

go to SOCO

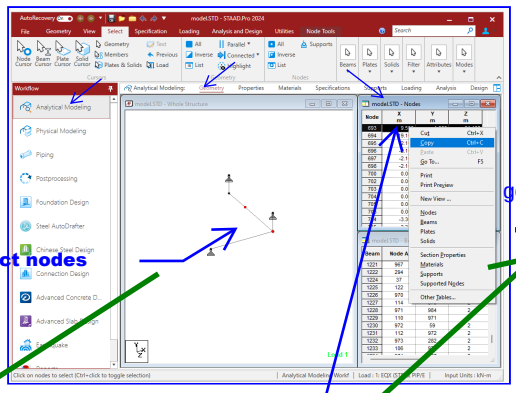
3. click 'get beam numbers'

beam numbers will appear

New API method- just select members and click



.. then get node numbers based on Staad selection to define beam ends (i, j):



4. select nodes

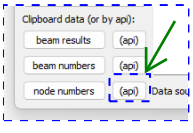
5. right click here and copy

go to SOCO

6. click 'get node numbers

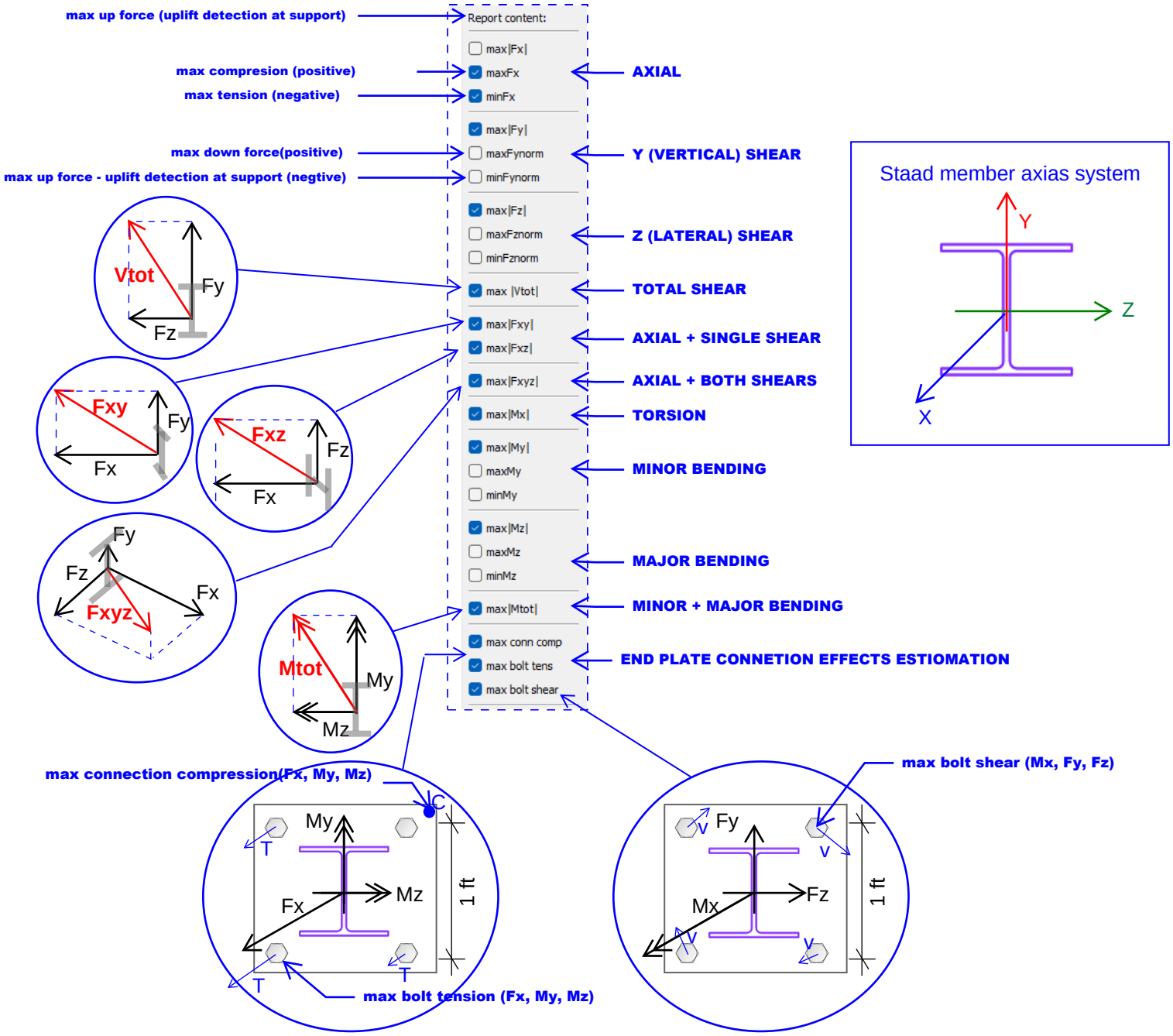
i and j will appear appropriately

New API method- just select members and click

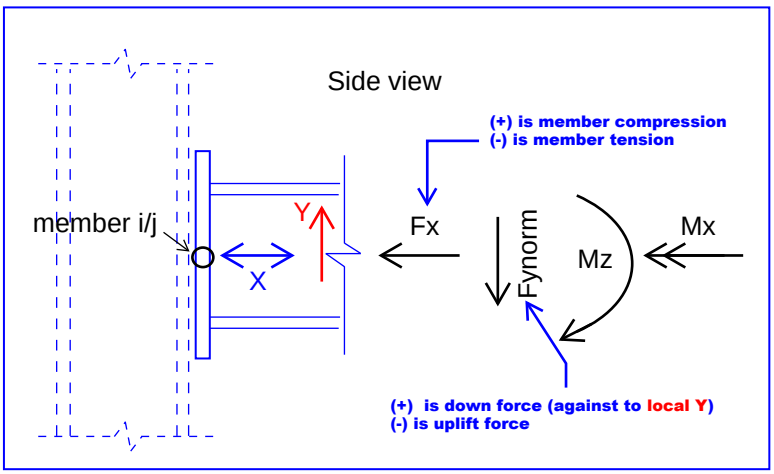


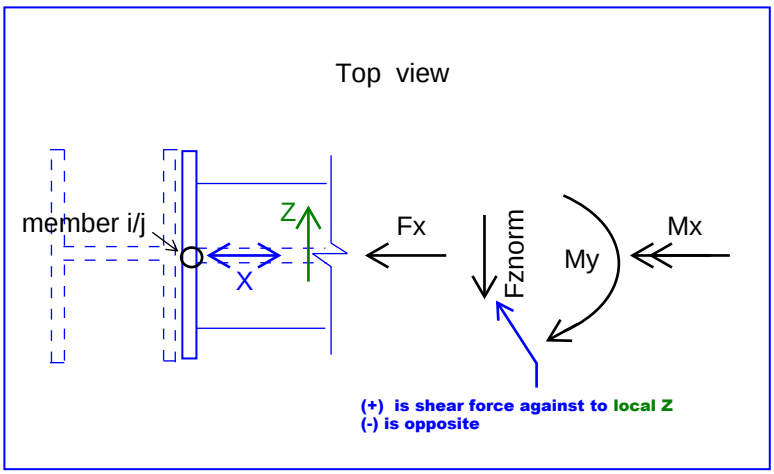
3. Decide what force criteria should be checked to find critical load cases

There are several options that can be selected to get output report, see below.

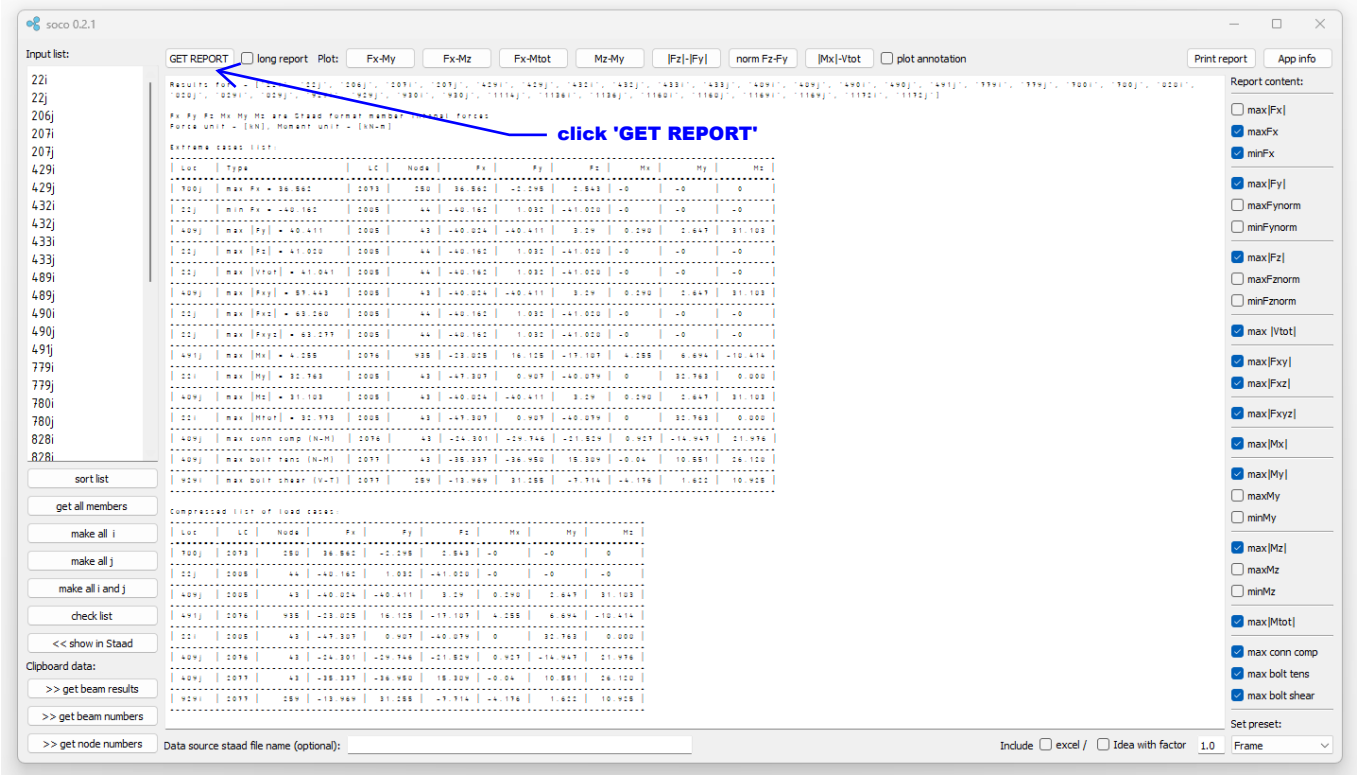


Force sign convention



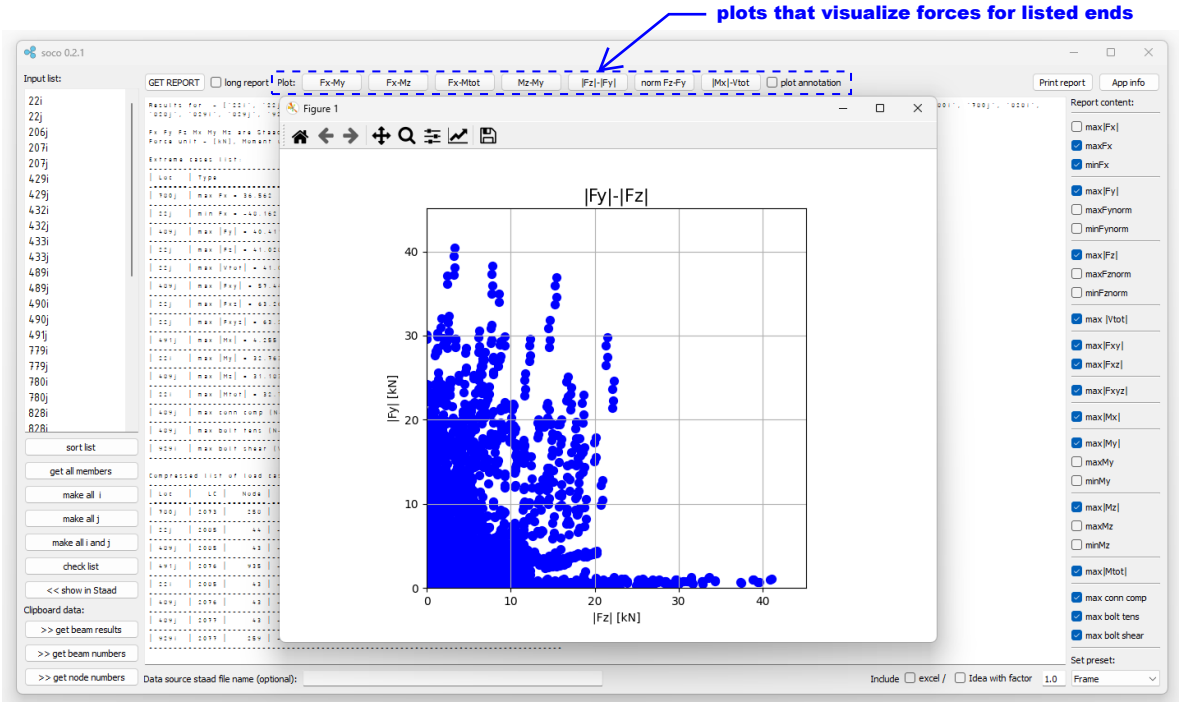


4. Generating output report

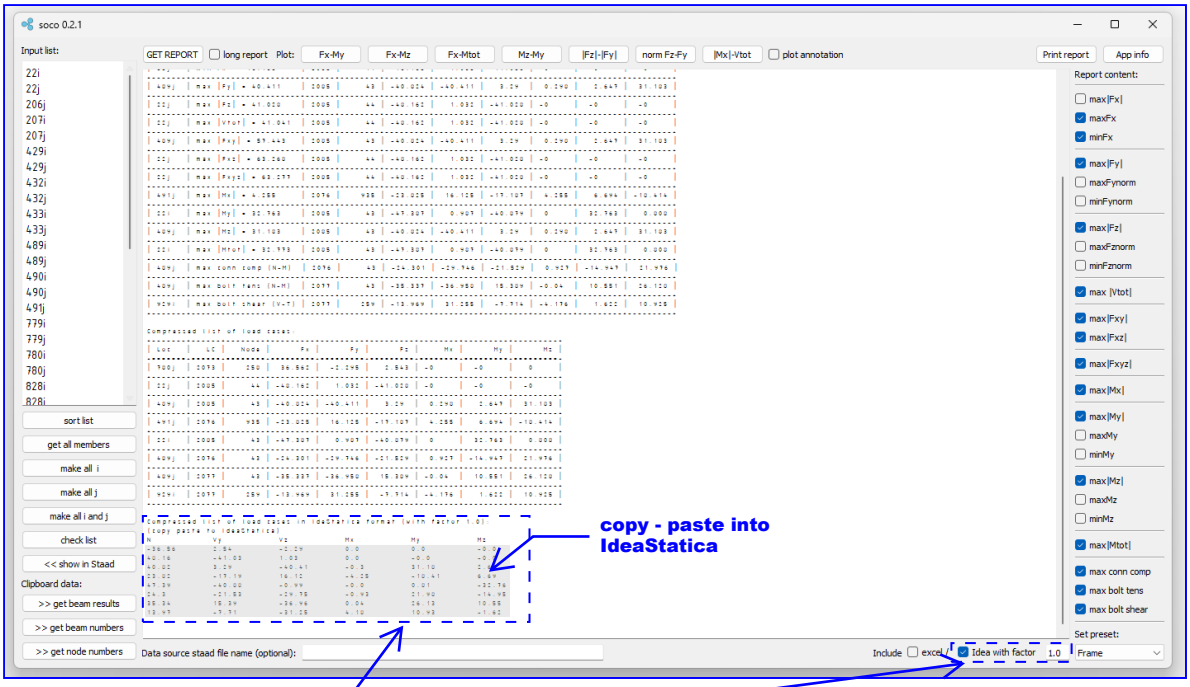


Additional features

Plots

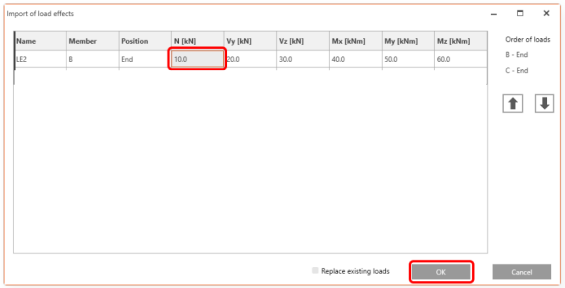


Transfering force to IdeaStatica

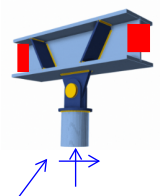


IdeaStatica input

Copy area as show above.
In IDEA StatiCa Connection, select the XLS Import command from the ribbon.
Finally, select the indicated cell below and Paste (Ctrl+V) the values.

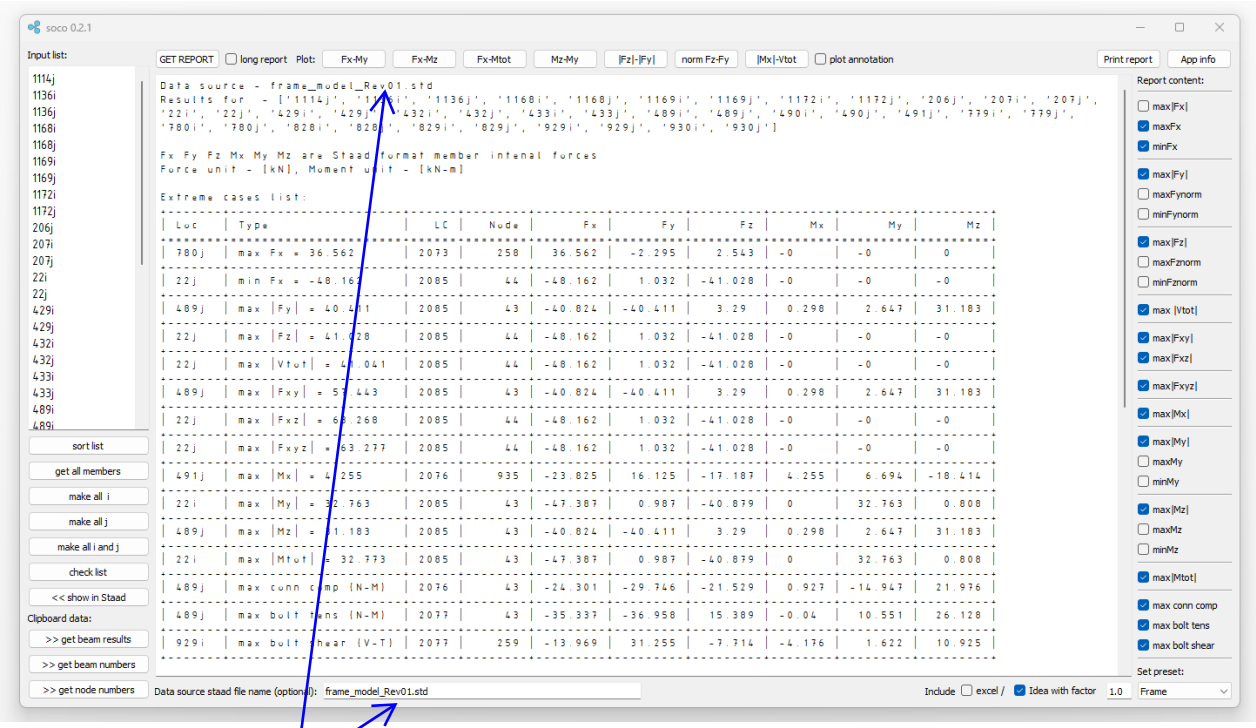


"One branch loaded" model example



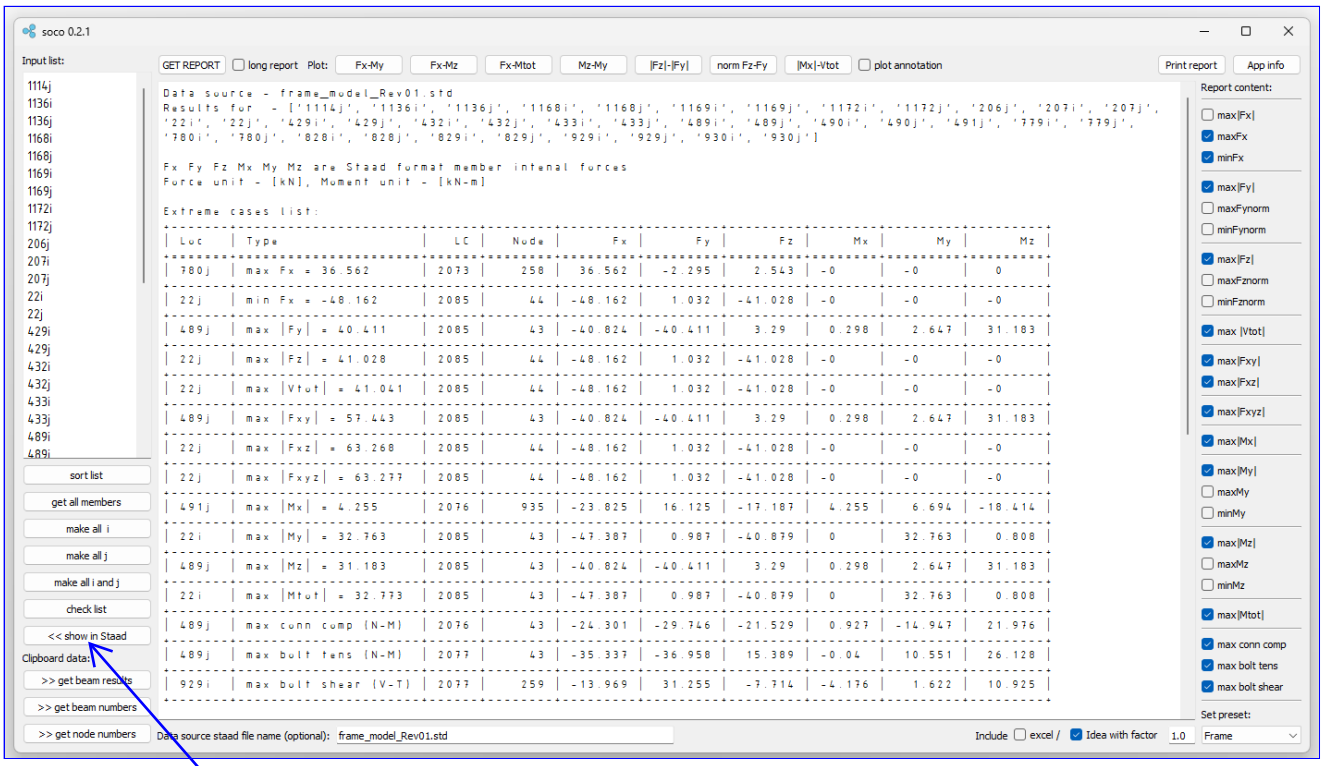
Note. This procedure is assumed to be used for IdeaStatic connection model with only one branch loaded.

Showing Staad model file name in report



optional Staad model name - if specified it will be shown at the top of report

Highlighting members and nodes in Staad



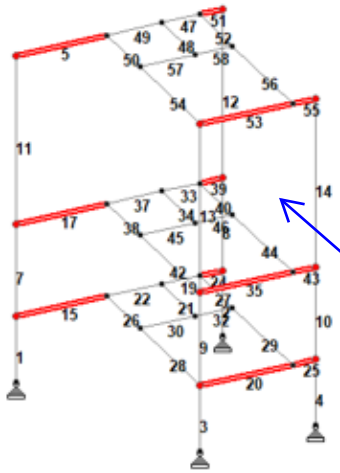
'show in staad' highlights members and nodes in Staad so you can see what members and ends are listed in Soco (make sure you have only one Staad instance open)

Showing soco output in calculation report

By default consider to insert Soco output in your report in the way shown below.

10.1 SHS200x200x10 beam to SHS200x200x10 post connection

calc report section title



show this snip from Staad, highlight beams and nodes that have been considered (you can use "show in staad" feature to highlight)

specify Staad Envelope or range of Load Cases that have be used to get Soco output

Factored internal forces (based on Staad "Beam End Forces" output data table for Envelope 3):

Results for - ['5i', '15i', '17i', '20j', '24j', '25j', '35j', '39j', '43j', '51j', '53i', '55j']

Fx Fy Fz Mx My Mz are Staad format member internal forces
Force unit - [kip], Moment unit - [kN-m]

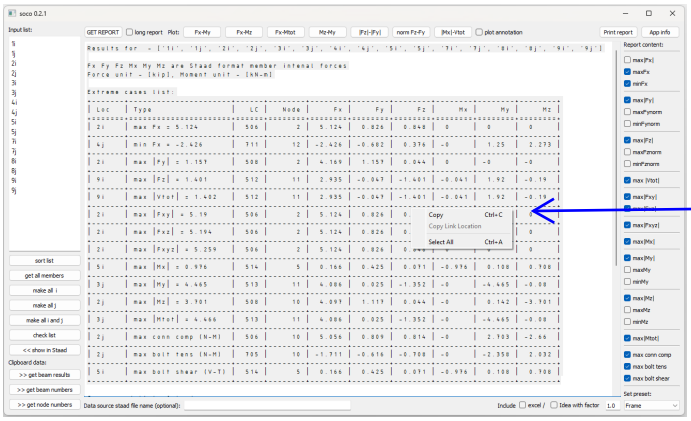
Extreme cases list:

Loc	Type	LC	Node	Fx	Fy	Fz	Mx	My	Mz
39j	max Fx = 0.551	502	14	0.551	-1.529	-0.227	1.808	-0.065	4.126
43j	min Fx = -0.41	509	16	-0.41	0.783	-0.016	-0.001	0.01	-2.47
24j	max |Fy| = 2.174	506	10	-0.061	-2.174	-0.033	3.958	0.202	4.47
39j	max |Fz| = 0.493	514	14	0.386	-1.524	-0.493	3.8	-0.117	2.198
24j	max |Vtot| = 2.174	506	10	-0.061	-2.174	-0.033	3.958	0.202	4.47
24j	max |Fxy| = 2.175	506	10	-0.06	-2.174	-0.033	3.958	0.202	4.47
39j	max |Fxz| = 0.627	506	14	0.496	-1.626	-0.383	3.157	-0.092	3.359
24j	max |Fxyz| = 2.175	506	10	-0.061	-2.174	-0.033	3.958	0.202	4.47
24j	max |Mx| = 4.917	514	10	-0.187	-1.972	-0.122	4.917	0.137	2.731
35j	max |My| = 0.817	509	15	-0.136	-1.016	-0.29	-0.128	-0.817	2.694
24j	max |Mz| = 5.623	502	10	0.06	-2.068	0.049	2.155	0.218	5.623
24j	max |Mtot| = 5.627	502	10	0.06	-2.068	0.049	2.155	0.218	5.623

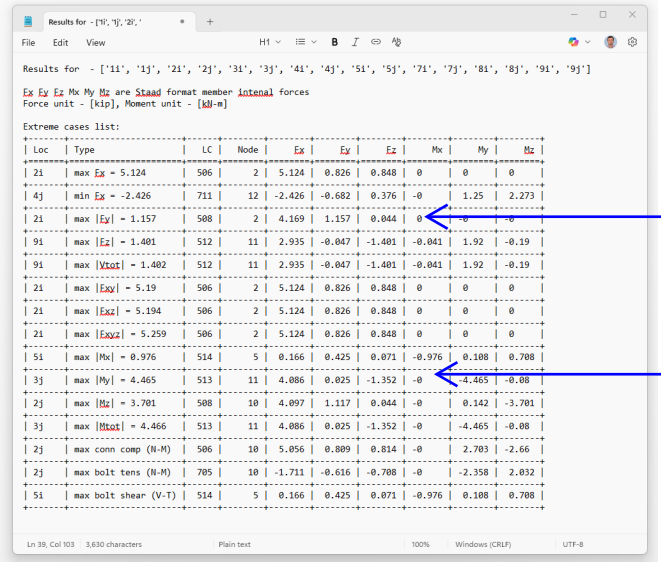
paste this as snip from soco

Getting Soco report as text in your Word calculation report

Instead of snip you can place Soco output as text in your calc report. As the Soco output text tables require monospace font follow the procedure shown below. (For some reason directly copy from Soco and paste into Word may not works correctly so Notepad stage is required)



1. select content, right click and "Copy"



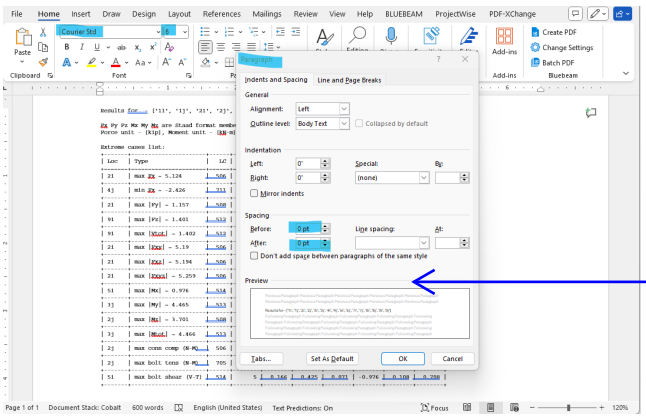
```
Results for - ['11', '1j', '21', '2j', '31', '3j', '41', '4j', '51', '5j', '71', '7j', '81', '8j', '91', '9j']

Fx Fy Fz Mx My Mz are Staad format member intenal forces
Force unit - [kip], Moment unit - [kN-m]

Extreme cases list:
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+
| Loc   | Type                    |  LC   | Node  |  Fx   |  Fy   |  Fz   |  Mx   |  My   |  Mz   |
+=======+=========================+=======+=======+=======+=======+=======+=======+=======+=======+
| 21    | max Fx = 5.124          | 506   |     2 | 5.124 | 0.826 | 0.848 | 0     | 0     | 0     |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 4j    | min Fx = -2.426         | 711   |    12 | -2.426 | -0.682 | 0.376 | -0    | 1.25  | 2.273 |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 21    | max |Fy| = 1.157       | 508   |     2 | 4.169 | 1.157 | 0.044 | 0     | -0    | -0    |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 91    | max |Fz| = 1.401       | 512   |    11 | 2.935 | -0.047 | -1.401 | -0.041 | 1.92  | -0.19 |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 91    | max |Vtot| = 1.402     | 512   |    11 | 2.935 | -0.047 | -1.401 | -0.041 | 1.92  | -0.19 |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 21    | max |Fxy| = 5.19       | 506   |     2 | 5.124 | 0.826 | 0.848 | 0     | 0     | 0     |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 21    | max |Fxz| = 5.194      | 506   |     2 | 5.124 | 0.826 | 0.848 | 0     | 0     | 0     |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 21    | max |Fxyz| = 5.259     | 506   |     2 | 5.124 | 0.826 | 0.848 | 0     | 0     | 0     |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 51    | max |Mx| = 0.976       | 514   |     5 | 0.166 | 0.425 | 0.071 | -0.976 | 0.108 | 0.708 |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 3j    | max |My| = 4.465       | 513   |    11 | 4.086 | 0.025 | -1.352 | -0    | -4.465 | -0.08 |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 2j    | max |Mz| = 3.701       | 508   |    10 | 4.097 | 1.117 | 0.044 | -0    | 0.142 | -3.701 |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 3j    | max |Mtot| = 4.466     | 513   |    11 | 4.086 | 0.025 | -1.352 | -0    | -4.465 | -0.08 |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 2j    | max conn comp (N-M)     | 506   |    10 | 5.056 | 0.809 | 0.814 | -0    | 2.703 | -2.66 |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 2j    | max bolt tens (N-M)     | 705   |    10 | -1.711 | -0.616 | -0.708 | -0    | -2.358 | 2.032 |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
| 51    | max bolt shear (V-T)    | 514   |     5 | 0.166 | 0.425 | 0.071 | -0.976 | 0.108 | 0.708 |
+-------+-------------------------+-------+-------+-------+-------+-------+-------+-------+-------+
```

2. Go to Notepad, right click and "Paste"

3. again select content, right click and "Copy"



4. Go to Word, right click and "Paste". Then set font size 6, font Courier Std, and Paragraph spacing 0.

- END OF TUTORIAL -

Project website:
https://github.com/lukaszlaba/soco

Download latest version from:
https://github.com/lukaszlaba/soco/releases